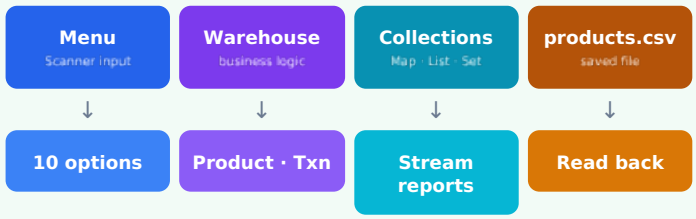
Warehouse Inventory & Reorder System

PROJECT BASED LEARNING FROM JAVA PROGRAMMING

Abstract

The **Warehouse Inventory & Reorder System** is a console application that keeps a live count of every product in a store. Stock changes through three movements: goods **brought in** from a supplier, goods **purchased** by a customer, and goods **returned**. A purchase larger than the stock in hand is rejected, so a count never falls below zero, and an **alert** is raised the moment stock reaches its **reorder point**. Built in **Java** using classes, inheritance, exceptions, collections and file handling.

System Architecture



The user picks an option, the Warehouse class updates the product and writes the movement to the ledger, the collections hold everything in memory, and the stock list is saved to a file and read back again.

MODULE-WISE IMPLEMENTATION

M1 Java Foundations & Program Model

- ✓ Product fields using primitive types and constants
- ✓ Bitwise flags for fragile and cold-storage items
- ✓ Ternary operator decides the LOW or OK status
- ✓ Typecasting for the average, printf for every report

```
stock report
CODE NAME QTY PRICE STATUS
P001 Pen 120 10.50 OK
P003 Glass Vial 40 18.75 OK
P004 Vaccine Box 18 950.00 LOW
```

M2 Control Flow

- ✓ Ten-option menu driven by a switch
- ✓ while and do-while loops for input retry
- ✓ Nested loops total the storage rack

M3 Methods, Recursion & Arrays

- ✓ returnGoods() overloaded two ways
- ✓ Recursive net effect over the ledger
- ✓ 1D statistics and a 3×4 rack matrix

M4 OOP & Exception Handling

- ✓ Product class with private fields
- ✓ Inward, Purchase, Return inherit Txn
- ✓ Custom exception on an oversized order

M5 Strings, Sorting & File I/O

- ✓ CSV split, join and StringBuilder
- ✓ Arrays.sort and binary search
- ✓ Save and reload, bad rows skipped

M6 Collections Framework & Streams

- ✓ HashMap for products, ArrayList for the ledger, HashSet for suppliers
- ✓ Lambdas and method references parameterise the reports
- ✓ filter, map, sorted and limit built into one pipeline
- ✓ Collectors groupingBy, counting and joining build the summaries

```
stream reports
Low stock items : [Vaccine Box]
Total value : 25811.25
Units/category : {STATIONERY=140, LAB=65}
Transactions : {RETURN=2, INWARD=2}
```

Conclusion & Future Scope

The project brings all six modules together in one working program, from primitive types and loops through to inheritance, exception handling, file I/O and streams. Stock stays correct through every movement, and the reorder alert makes it useful rather than only a coding exercise. In future it can be extended with a **database** in place of the CSV file, a **graphical interface**, barcode input, and **demand forecasting** to predict the reorder point instead of fixing it by hand.

JAVA SE **OOP & COLLECTIONS** **FILE HANDLING** **STREAMS & LAMBDAS** **EXCEPTIONS**